

CH5 Special Discrete Distributions

Bernoulli and binomial R.V.

Definition

X is a Bernoulli trial (or Bernoulli random variable) with parameter p if sample space $S=\{s, f\}$, where s is called a success and f a failure and $X(s)=1$, $X(f)=0$.

The probability mass function of X is

$$p(x) = \begin{cases} p & x = 1 \\ 1 - p \equiv q & x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times P(X = 0) + 1 \times P(X = 1) = p$$

$$E(X^2) = 0^2 \times P(X = 0) + 1^2 \times P(X = 1) = p$$

$$Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p)$$

Definition

$$X \sim B(n, p)$$

X : The number of success

n : Perform n independently event

p : The probability of success

The p.m.f. of $X \sim B(n, p)$

$$p(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & \text{if } x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases}$$

If X is a binomial random variable with parameters $B(n, p)$, we can obtain

$$\sum_{x=0}^n p(x) = \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} = [p + (1 - p)]^n = 1^n = 1$$

Note: Multinomial theorem is used

Theorem

If X is a binomial random variable with parameter $B(n, p)$, then as k goes from 0 to n , $P(X=k)$ first increases monotonically and then decreases monotonically, reaching its maximum value when k is the largest integer less than or equal to $(n+1)p$.

假設我們進行 $n=100$ 次試驗，每次成功的機率 $p=0.665$ 。 $(n+1)p = 101 \times 0.665 = 67.165$

根據定義， k 是小於或等於 67.165 的最大整數，所以 $k=67$

Ex 5.2 A restaurant serves 8 entrees of fish, 12 of beef, and 10 of poultry.

If customers select from these entrees randomly, what is the probability that 2 of the next four customers order fish entrees?

$$X(4, \frac{8}{30})$$

$$P(X=2) = \binom{4}{2} \left(\frac{8}{30}\right)^2 \left(\frac{22}{30}\right)^{4-2}$$

Ex 5.2 The probability of a bit error in a communication line is 0.001. Find the probability that a block of 10000 bits contains two or more errors?

$$X(10000, 0.001)$$

$$1 - P(X=0) - P(X=1)$$

Ex Consider a jury trial in which it takes 8 of the 12 jurors to convict; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. Assume that each of the jurors makes the right decision with probability p , while g represents the probability that the defendant is guilty. What is the probability that the jury makes a right decision?

情況 1: (有罪) $\rightarrow$ 至少 8 人做出正確判決

$$P(A|g) = P(A|g) \cdot P(g) = \sum_{k=8}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g)$$

情況 2: (無罪) $\rightarrow$ 至少 5 人判無罪

$$P(A|g^c) = P(A|g^c) \cdot P(g^c) = \sum_{k=5}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g^c)$$

Mean of Binomial random variable

If X is a binomial random variable

$$E(X) = np$$

$$E[X(X-1)] = n^2 p^2 - np^2$$

$$E(X^2) = n^2 p^2 - np^2 + np$$

$$Var(X) = E(X^2) - (E(X))^2 = np(1-p)$$

